

Jason Naradowsky

jason.narad@gmail.com

narad.github.io

Shinjuku, Tokyo • GitHub • LinkedIn

Research Interests

My research spans language, audio, simulation, and world models. I am particularly interested in uncovering the latent principles underlying effective prediction and coordination, and computational models of creativity and open-endedness.

Work Experience

- Apr. 2026–Present **Research Scientist**, SoftBank Intuitions, Tokyo
Creative Vision team
- Developing Genie-style world models for games, including latent action and action-conditioned dynamics models.
 - Exploring architectural variations with a focus on responsive control and real-time interaction.
 - Designing exploration agents to collect diverse experience from game environments for world-model training.
- 2025–2026 **Founder**, Independent venture, Tokyo
- Developed software for music production and real-time audio processing, including neural audio models, data capture and training pipelines, and low-latency inference.
- 2022–2024 **Senior Researcher / Technical Strategy**, Square Enix: AI & Arts Alchemy, Tokyo
- Led the design and development of in-house ML systems for real-time virtual avatars, spanning language, audio, and animation under strict latency and deployment constraints.
 - Designed and fine-tuned bilingual (Japanese/English) LLM pipelines for local deployment, balancing model size, responsiveness, and behavioral consistency in interactive settings.
 - Developed audio-driven facial animation models and removed inference bottlenecks, reducing end-to-end latency by a factor of approximately 20 while maintaining perceptual fidelity.
 - Built speech synthesis and audio analysis models, balancing quality, controllability, and inference cost through user-facing evaluation.
 - Developed models for inferring emotional state and communicative intent from user audio and integrated them into interactive character systems.
- 2020–Present **Project Assistant Professor**, Miyao Lab, University of Tokyo, Tokyo
- Led a research group focused on dialogue generation, emergent communication, and controllable behavior in LLM-based agents.
 - Published work at ACL and EMNLP on long-term dialogue consistency, controllable generation, and hybrid neural-structured systems.
 - Investigated methods for analyzing and evaluating LLM behavior beyond surface metrics, including internal state tracking, multi-hop reasoning, and behavioral consistency over time (LLM-JP grant).

- 2018–2022 **Researcher**, Preferred Networks, Tokyo
- Conducted research on neural sequence modeling across robotics, singing voice synthesis, and generative audio, with an emphasis on representation learning and controllability.
 - Explored early DDSP-style approaches to audio modeling, including factorized representations for pitch, timbre, and control.
 - Mentored interns on end-to-end research projects, resulting in publications at ICASSP and DAFx.
- 2017–2018 **Associate Research Scientist**, Johns Hopkins University (HLTCOE), Baltimore
- Developed neural-CRF and transformer-based models for named entity recognition, examining trade-offs between structured inference and end-to-end learning.
 - Conducted research on entailment, bias, and morphologically rich NLP, with a focus on how modeling assumptions affect generalization.
- 2016–2017 **Senior Research Associate**, University of Cambridge, Cambridge
- Conducted research in multilingual NLP, including methods for altering and probing embedding-space representations across languages.
- 2015–2016 **Co-founder**, Bloomsbury AI, London
- Co-founded an NLP startup and developed tools for applying then-emerging deep learning methods to natural language processing.
- 2014–2016 **Senior Research Associate**, University College London, London
- Developed state-of-the-art models for distantly supervised event extraction using pointer networks.
 - Applied α -bound theory to imitation learning for AMR parsing and coreference, analyzing stability and exploration under structured constraints.
- 2010 **Research Intern**, Microsoft Research, Redmond
- Conducted research on morpheme-based alignment models for machine translation.
 - Developed joint morpheme segmentation and alignment models for Arabic using context-rich probabilistic models.
- 2008 **Google Summer of Code**, NLTK
- Implemented dependency parsers and supporting interfaces for the Natural Language Toolkit.

Education

- 2008–2014 **PhD in Computer Science**, University of Massachusetts Amherst, USA
- Dual degree program with Macquarie University.
- Advisor: David A. Smith
 - Certificate in Cognitive Science
- 2011–2014 **PhD in Computer Science**, Macquarie University, Australia
- Advisor: Mark Johnson
- 2008 **MSc, Artificial Intelligence**, University of Edinburgh, UK
- Thesis: *Improving Morphology Induction with Phonological Rules*
 - Advisor: Sharon Goldwater
- 2007 **MS, Computational Linguistics**, SUNY Buffalo, USA
- Thesis: *The Effect of Frequencies and Unseen Events on Parser Portability*
 - Advisor: Doug Roland
- 2006 **MA, Human–Computer Interaction**, SUNY Oswego, USA
- Thesis: *Neural Networks for Automated Design Evaluation*
 - Advisor: Craig Graci

- 2001–2005 **BS, Computer Science & BA, Linguistics**, SUNY Oswego, USA
- Specialization: Artificial Intelligence, with Honors
 - Minor: Cognitive Science
 - Thesis: *Baroque Music Generation Using Genetic Algorithms*

Doctoral Thesis

- Title *Learning with Joint Inference and Latent Linguistic Structure in Graphical Models*
- Supervisors David A. Smith and Mark Johnson
- Description Developed a modeling framework for constructing joint factor graph models of NLP problems, marginalizing over latent syntactic representations subject to combinatorial constraints to produce task-specific syntactic distributions.

Teaching Experience

- 2009 Grader, CMPSCI 585: Introduction to NLP, UMass Amherst

Advising

Master's Students

- 2016 Chris Loy (UCL), *Deep Hierarchical Architectures for Polyphonic Music Transcription*
- 2015 James Goodman (UCL), *Semantic Parsing from English to AMR Using Imitation Learning*

Undergraduate Committees

- 2013 Elias Zeidan (Marlboro College)

Publications

Journal Publications

- [1] Daniela Gerz, Ivan Vulić, Eduardo Ponti, Jason Naradowsky, Roi Reichart, and Anna Korhonen. Language modeling for morphologically rich languages: Character-aware modeling for word-level prediction. *Transactions of the Association for Computational Linguistics*, 2018.

Refereed Conference Proceedings

- [1] Zhan Shen, Xiaotian Wang, Jason Naradowsky, and Yusuke Miyao. Self-evolving optimization of agentic systems for machine translation. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026.
- [2] Kunhang Li, Jason Naradowsky, Yansong Feng, and Yusuke Miyao. How much do large language models know about human motion? A case study in 3D avatar control. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 13878–13921, Suzhou, China, November 2025. Association for Computational Linguistics.

- [3] Qiang Zhang, Jason Naradowsky, and Yusuke Miyao. Self-emotion blended dialogue generation in social simulation agents. In Tatsuya Kawahara, Vera Demberg, Stefan Ultes, Koji Inoue, Shikib Mehri, David Howcroft, and Kazunori Komatani, editors, *Proceedings of the 25th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 228–247, Kyoto, Japan, September 2024. Association for Computational Linguistics.
- [4] Shunsuke Kando, Yusuke Miyao, Jason Naradowsky, and Shinnosuke Takamichi. Textless dependency parsing by labeled sequence prediction. In *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [5] Daichi Kato, Ryo Ueda, Jason Naradowsky, and Yusuke Miyao. Emergent communication with stack-based agents. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society (CogSci)*, Rotterdam, The Netherlands, 2024.
- [6] Qiang Zhang, Jason Naradowsky, and Yusuke Miyao. Mind the gap between conversations for improved long-term dialogue generation. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10735–10762, Singapore, December 2023. Association for Computational Linguistics.
- [7] Qiang Zhang, Jason Naradowsky, and Yusuke Miyao. Ask an expert: Leveraging language models to improve strategic reasoning in goal-oriented dialogue models. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 6665–6694, Toronto, Canada, July 2023. Association for Computational Linguistics.
- [8] Wenjie Zhong, Jason Naradowsky, Hiroya Takamura, Ichiro Kobayashi, and Yusuke Miyao. Fiction-writing mode: An effective control for human-machine collaborative writing. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1752–1765, Dubrovnik, Croatia, May 2023. Association for Computational Linguistics.
- [9] Ryokan Ri, Ryo Ueda, and Jason Naradowsky. Emergent communication with attention. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (CogSci)*, Sydney, Australia, 2023.
- [10] Qiang Zhang, Jason Naradowsky, and Yusuke Miyao. Rethinking offensive text detection as a multi-hop reasoning problem. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3888–3905, Dublin, Ireland, May 2022. Association for Computational Linguistics.
- [11] Jason Naradowsky. Amp-Space: A large-scale dataset for fine-grained timbre transformation. *Proceedings of the 24th International Conference on Digital Audio Effects (DAFx-21)*, Vienna, Austria, 2021.
- [12] Jason Naradowsky, Xuan Zhang, and Kevin Duh. Machine translation system selection from bandit feedback. In *AMTA*, 2020.
- [13] David Samuel, Aditya Ganeshan, and Jason Naradowsky. Meta-learning extractors for music source separation. In *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020.

- [14] Rachel Rudinger, Jason Naradowsky, Brian Leonard, and Benjamin Van Durme. Gender bias in coreference resolution. In *NAACL*, 2018.
- [15] Lawrence Wolf-Sonkin, Jason Naradowsky, Sebastian J. Mielke, and Ryan Cotterell. A structured variational autoencoder for contextual morphological inflection. In *Association for Computational Linguistics (ACL)*, 2018.
- [16] Matko Bošnjak, Tim Rocktäschel, Jason Naradowsky, and Sebastian Riedel. Programming with a differentiable Forth interpreter. In *International Conference on Machine Learning (ICML)*, 2017.
- [17] Lucas Sterckx, Jason Naradowsky, Bill Byrne, Thomas Demeester, and Chris Develder. Break it down for me: A study in automated lyric annotation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2017.
- [18] James Goodman, Andreas Vlachos, and Jason Naradowsky. Noise reduction and targeted exploration in imitation learning for abstract meaning representation parsing. In *Association for Computational Linguistics (ACL)*, 2016.
- [19] James Goodman, Andreas Vlachos, and Jason Naradowsky. UCL+Sheffield at SemEval-2016 task 8: Imitation learning for AMR parsing with an alpha-bound. In *Proceedings of the 10th International Workshop on Semantic Evaluation*, 2016.
- [20] Jason Naradowsky, Sebastian Riedel, and David Smith. Improving NLP through marginalization of hidden syntactic structure. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2012.
- [21] Jason Naradowsky, Tim Vieira, and David A. Smith. Grammarless parsing for joint inference. In *24th International Conference on Computational Linguistics (COLING)*, Mumbai, India, 2012.
- [22] John Lee, Jason Naradowsky, and David Smith. A discriminative model for joint morphological disambiguation and dependency parsing. In *Association for Computational Linguistics (ACL)*, 2011.
- [23] Jason Naradowsky and Kristina Toutanova. Unsupervised bilingual morpheme segmentation and alignment with context-rich hidden semi-markov models. In *Association for Computational Linguistics (ACL)*, 2011.
- [24] David Mimno, Hanna Wallach, Jason Naradowsky, David Smith, and Andrew McCallum. Polylingual topic models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2009.
- [25] Jason Naradowsky and Sharon Goldwater. Improving morphology induction by learning spelling rules. In *International Joint Conference on AI (IJCAI)*, pages 1531–1537, 2009.

Workshop Proceedings

- [1] Pratik Sutar, Jason Naradowsky, and Yusuke Miyao. "Does it Chug?" Towards a data-driven understanding of guitar tone description. In *NLP4MUSA*, 2024.

- [2] Takuma Yoneda, Matthew Walter, and Jason Naradowsky. Pow-Wow: A dataset and study on collaborative communication in Pommerman. In *ICML Workshop on Language in Reinforcement Learning (LaReL)*, 2020.
- [3] Alexander Cowen-Rivers and Jason Naradowsky. Emergent communication with world models. In *NeurIPS Workshop on Emergent Communication*, 2019.
- [4] Fabrice Matulic, Yuta Kikuchi, and Jason Naradowsky. Enabling customer-driven learning and customisation processes for ML-based domestic robots. In *CHI Human-Centered Machine Learning Perspectives Workshop*, 2019.
- [5] Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. Hypothesis only baselines in natural language inference. In *Proceedings of the 7th Joint Conference on Lexical and Computational Semantics, *SEM 2018*, New Orleans, Louisiana, June 2018. Association for Computational Linguistics.
- [6] Zhengyan Gao, Taizan Yonetsuji, Tatsuya Takamura, Toru Matsuoka, and Jason Naradowsky. Automatic illumination effects for 2D characters. In *NeurIPS Workshop on Machine Learning for Creativity and Design*, 2018.
- [7] Naoyuki Kanda, Rintaro Ikeshita, Shota Horiguchi, Yusuke Fujita, Kenji Nagamatsu, Xiaofei Wang, Vimal Manohar, Nelson Enrique Yalta Soplin, Matthew Maciejewski, Szu-Jui Chen, Aswin Shanmugam Subramanian, Ruizhi Li, Zhiqi Wang, Jason Naradowsky, L. Paola Garcia-Perera, and Gregory Sell. The Hitachi/JHU CHiME-5 system: Advances in speech recognition for everyday home environments using multiple microphone arrays. In *The 5th International Workshop on Speech Processing in Everyday Environments (CHiME)*, 2018.
- [8] Azumi Maekawa, Ayaka Kume, Hironori Yoshida, Jun Hatori, Jason Naradowsky, and Shunta Saito. Improvised robotic design with found objects. In *NeurIPS Workshop on Machine Learning for Creativity and Design*, 2018.
- [9] Jason Naradowsky and Sebastian Riedel. Modeling exclusion with a differentiable factor graph constraint. In *Workshop on Deep Structured Prediction, International Conference on Machine Learning (ICML)*, 2017.
- [10] Matko Bošnjak, Tim Rocktäschel, Jason Naradowsky, and Sebastian Riedel. A neural Forth abstract machine. In *Neural Abstract Machines & Program Induction (NAMPI)*, Barcelona, 2016.
- [11] Matko Bošnjak, Tim Rocktäschel, Jason Naradowsky, and Sebastian Riedel. A neural Forth abstract machine. In *RNN Symposium*, Barcelona, 2016.
- [12] Jason Naradowsky, Joe Pater, and David Smith. Feature induction for online constraint-based phonology acquisition. In *The Northeast Computational Phonology Workshop (NECPhon)*, New Haven, Connecticut, 2011.
- [13] Jason Naradowsky, Joe Pater, David Smith, and Robert Staubs. Learning hidden metrical structure with a log-linear model of grammar. In *Computational Modelling of Sound Pattern Acquisition*, pages 59–60, Edmonton, 2010.

- [14] David Mimno, Hanna Wallach, Limin Yao, and Jason Naradowsky. Polylingual topic models. In *The Learning Workshop (Snowbird)*, Clearwater, Florida, 2009.

Demo Proceedings

- [1] Sameer Singh, Tim Rocktäschel, Luke Hewitt, Jason Naradowsky, and Sebastian Riedel. WOLFE: An NLP-friendly Declarative Machine Learning Stack. In *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2015.

Tutorials

- 2015 Matrix and Tensor Factorization Methods for Natural Language Processing, *ACL 2015*

Invited Talks

- July 16, 2026 *From Pixels to Playable Worlds: Turning AI's World Knowledge into Real Games*, Runway Meetup Tokyo: Creative Vision and the Future of Interactive AI
- 2023 *Rethinking Offensive Text Detection as a Multi-Hop Reasoning Problem*, NAIST
- 2023 *The Future of NLP in Game Design: The Next Revolution in Gaming*, NAIST
- 2017 *Eliminating Misinformation in Machine Reading with Aggregation*, Kyoto University
- 2016 *Distantly Supervised Event Extraction with Pointer Networks*, Tokyo Metropolitan University
- 2016 *Computers That Read: Uncovering the Structure of Language with Deep Learning*, Osaka University
- 2016 *Distantly Supervised Event Extraction with Pointer Networks*, NAIST
- 2016 *Artificial Intelligence: A Rationalist Perspective on the Past and Future of AI*, Embassy of Japan, London
- 2015 *Deep Sequence Models, Multimodality & Conversational Agents*, Osaka University
- 2015 *Learning Latent Syntactic Representations with Joint Models*, Xerox Research Center, Grenoble
- 2015 *Learning Latent Syntactic Representations with Joint Models*, Cambridge University

Professional Service

Area Chair

- 2023–2026 ACL, ARR
- 2019 ACL (Machine Learning Track)

Program Committee / Reviewer

- 2020–2023 ACL, EMNLP, ICLR, ICML, NeurIPS
- 2019 AKBC, CoNLL, EMNLP, ICML, NAACL, NeurIPS, Repl4NLP, Typ-NLP
- 2010–2018 ACL, CoNLL, EMNLP, NAACL, NeurIPS, AAI, COLING

Journal Reviewer

- 2016–Present Transactions of the Association for Computational Linguistics (TACL)
2015 Transactions on Audio, Speech and Language Processing (T-ASL)

Organization

- 2016 Co-organizer, AI4Exams (with Yusuke Miyao & Sebastian Riedel)

Awards and Achievements

- 2020 Best Reviewer, NeurIPS (Top 10%) and ICML
2018 Best Paper Award, *SEM
2015 Daiwa Foundation Small Grant Award
2014 Best Reviewer, ACL
2012 NSF East Asia and Pacific Summer Institute (EAPSI) Fellowship
2012 Best Reviewer, EMNLP
2011 Cotutelle International Macquarie University Research Scholarship (iMQRES)
2005 Oebele Van Dyk Outstanding Senior in Computer Science Award
2001–2005 Presidential Scholarship, SUNY Oswego

Technical Skills

- Languages English (native), Japanese (beginner), Latin (reading)
Programming Python, Scala, Java, Ruby, Clojure, Common Lisp
ML Frameworks PyTorch, TensorFlow, JAX; ONNX for model interchange
Game Engines Unreal Engine, Unity, Godot
AI/ML World Models, Latent Action Models, LLMs, Dialogue & Agent Systems,
Real-time Inference, Generative Audio, Speech Synthesis, Facial Animation,
NLP, Structured Prediction / Joint Inference

Additional Information

- Citizenship USA